09: Intro to Multiple Regression

Stat 230 - Spring 2026

```r
library(ggformula)
library(dplyr)
library(broom)
library(GGally)

supervisor <- read.csv("http://aluby.github.io/stat230/data/supervisor.csv")
```

1 Data

- In an effort to understand the important aspects of a satisfactory supervisor, clerical employees at a large financial organization were asked to rate their immediate supervisor.
- The survey questions were designed to measure overall performance of the supervisor, as well as six additional characteristics.
- Employees were asked to rate the following statements on a scale from 0 to 100

  (0 = "completely disagree", 100 = "completely agree")

Note: this data comes from the textbook

Variable	Description
`rating`	Overall rating of supervisor performance
`complaints`	Score for "Your supervisor handles employee complaints appropriately."
`privileges`	Score for "Your supervisor allows special privileges."
`learn`	Score for "Your supervisor provides opportunities to learn new things."
`raises`	Score for "Your supervisor bases raises on performance."
`critical`	Score for "Your supervisor is too critical of poor performance."

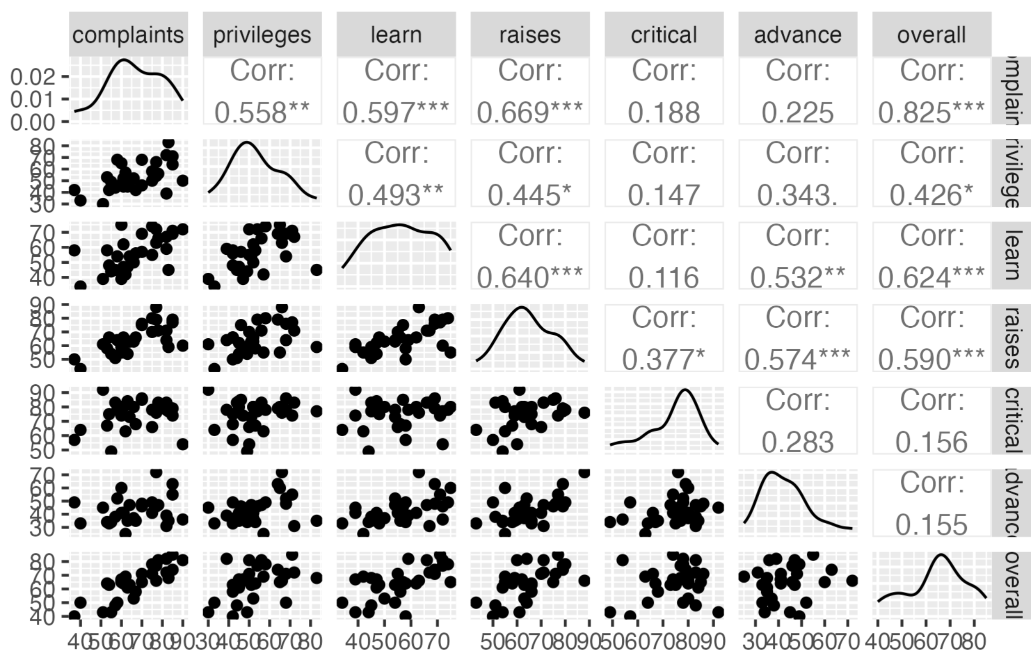
Variable	Description
advance	Score for "I am not satisfied with the rate I am advancing in the company?"

Before looking at the data, how do you think each predictor variable will be related to `rating`? What do you think will have the strongest relationship? The weakest?

1.1 EDA: Making a pairs plot

We will use the `ggpairs()` function in the {GGally} package to make the pairs plot. By default, it will include every variable in the dataset. The code below firsts `select()`s the columns we want to include (all of them). I have also moved the response variable to be the last in the list, so that it shows up on the Y axis.

```
supervisor |>
  select(complaints, privileges, learn, raises, critical, advance, overall) |>
  ggpairs()
```



- Which predictor variables appear to be most closely related to `overall`?
- Which predictor variables appear to be most closely related to *each other*?

2 SLR

In the code chunk below, choose what you think will be the best *single* predictor and fit an SLR model. Interpret the intercept and slope coefficient and report the R^2 value.

```
# your code here
```

3 MLR

In the code chunk below, use all predictors and fit an MLR model. Interpret the intercept and slope coefficients and report the R^2 value.

```
# your code here
```

4 Centered/Standardized predictors

Decide whether you think standardized or centered predictors would be more useful here, then fit the MLR model and provide interpretations of the intercept and slope coefficients. Did R^2 change?

```
# your code here
```

5 Predicting responses

We can use most our SLR tools for MLR. For example, here's some code I could use to calculate predictions.

```
predict(bikes_lm, newdata = data.frame(temp_actual = 90, windspeed = 6))
```

- Use this as starter code to predict the overall manager score for a manager with scores of 50% for all predictors
- What if the manager from the previous question decides to try to improve her `critical` score by 10 points (so 40% instead of 50%) and her `complaints` score by 10 points (so 60% instead of 50%). How does that change the prediction?

```
# your code here
```

5.1 Function quick reference

The following table summarizes the functions we learned today:

Function	Purpose
`lm(formula, data)`	Fit a MLR model
`predict(model, newdata, interval, level, se.fit)`	Make predictions and calculate intervals
`data |> select(vars) |> ggpairs()`	Make pairs plot for selected variables